

2(a)(i)	<i>any two from</i> each polypeptide is a (left-handed), helix / helical shape ; R alpha helix (three) polypeptides are tightly, coiled / wound (around each other) / AW ; (three polypeptides form a) <u>triple helix</u> ; I bonding	2
2(a)(ii)	<i>any three from</i> <i>collagen molecules arranged</i> 1 in parallel ; 2 staggered / ends are not aligned / AW ; <i>molecules held together by – max 2</i> 3 covalent, bonds / links ; I hydrogen bonds R if part of a list that includes other bonds in proteins 4 between R-groups of amino acids on different molecules ; A side chains 5 AVP ; e.g. covalent bonds between lysine / hydroxylysine (residues)	3
2(b)(i)	provides (high tensile) strength to, help keep trachea open / support trachea ; A provides strength to prevent trachea collapsing allows flexibility for, breathing / change of diameter of trachea / bending of trachea / AW ;	2
2(b)(ii)	<i>any one from:</i> (inhalation) allow, stretching / expansion, without, rupture / AW ; A prevent overstretching elastic fibres recoil to help, expel air / exhalation / expiration ; R to empty	1
2(c)(i)	(gly)-pro-met-gly-pro-arg-gly-pro-pro-gly ;	1

Question	Answer	Marks
2(c)(ii)	any three from: <i>allow gly for glycine</i> 1 glycine is every third amino acid ; 2 glycine, allows tight folding for the polypeptides / makes it easy for polypeptides to fit closely together / AW ; A compactness A glycine allows the turning points (of the polypeptide) 3 glycine has <u>smallest</u> R-group ; A glycine is the <u>smallest</u> (amino acid) / glycine has H as R-group or side chain 4 NH in glycine can form hydrogen bonds (with carboxyl group on another amino acid residue) 5 (so) allows many hydrogen bonds between polypeptides ; 6 AVP ; ala / pro, also has a small R-group / AW	3
2(c)(iii)	P <i>deletion of first nucleotide pair</i> changes the, sequence / order, of amino acids (in primary structure) ; I shorter A (GTC CAA TGG GTC CCC GAG GTC CCC CAG GT_) val gln trp val pro glu val pro gln (val) Q <i>substitution in first triplet</i> (GGT to TGT in non-transcribed strand) cys (instead of gly) is first amino acid, <u>and</u> no change to the rest (of, sequence / order, of amino acids) ; A <u>only</u> first amino acid is different	2